

Department of Computer Science & Engineering
Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

PROJECT CENTRIC LEARNING (PCL) PHASE 2

		PROJECT TEAM NO.		CoEPL-0177
PROJECT TITLE		AI-Powered Smart Reverse Vending Machine For Automated Waste Classification and Reward System.		
TEAM MEMBERS				
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DETAILS OF THE PROJECT GUIDE				
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DESIGNATION		Associate Professor		
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DOMAIN OF THE PROJECT				
Artificial Intelligence, Machine Learning, Computer Vision, Full Stack Development				
Scope of the Project				
1. Automated waste Classification using AI-based image recognition				
2. Encouraging recycling behavior using reward-based smart RVM systems				
3. Real time waste classification through web-based deployment architecture				
4. Scalable architecture for future hardware & IoT integration.				

PROJECT GUIDE

PROJECT COORDINATOR

Existing System:-

Traditional waste segregation systems rely heavily on manual sorting, which is inefficient, time consuming, and prone to human error. Existing reverse vending machine are expensive & limited to pre defined recyclable materials.

Problem Identification

Improper waste segregation leads to environmental pollution and inefficient recycling processes. There is a need for a low cost intelligent system capable of automatically classifying recyclable waste materials with high accuracy.

Proposed System:-

The proposed Smart Reverse Vending Machine uses an AI-powered image classification model to identify waste categories such as Plastic, Paper, Glass, Metal, Hazardous and Organic waste. The system integrates a flask backend, Dockerized machine learning inference engine, and responsive frontend interface to provide real time classification and simulation.

No. of Modules

Sl. No.	Name of the Module	Remarks
1	Dataset Collection & Preprocessing	

2	Model Training & Validation	
3	Backend API Development	
4	Frontend Interface	
5	Docker Development	
6	Result Visualization & Released system.	

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